

2.2.1 The heart and monitoring heart function

Learning outcomes	Additional guidance
<i>Learners should be able to demonstrate and apply their knowledge and understanding of:</i>	
(a) the need for a mass transport system in mammals	$Ratio = \frac{Surface\ Area}{Volume}$ To include references to a high basal metabolic rate, being multicellular and the significance of surface area to volume ratio. <i>M0.3, M4.1</i>
(b) (i) the internal and external structure of the mammalian heart (ii) the examination, dissection and drawing of the mammalian heart	PAG2 HSW3, HSW4, HSW5, HSW6, HSW8
(c) the cardiac cycle	To include the role of the valves and the pressure changes occurring in the heart and associated vessels.
(d) how heart action is initiated and co-ordinated	To include the roles of the sino-atrial node (SAN), atrio-ventricular node (AVN), purkyne tissue and the myogenic nature of cardiac muscle (no detail on hormonal and nervous control is required at AS level but at A level this is covered in 5.3.1(b)).

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| <p>(e) practical investigation(s) into the factors affecting heart rate</p> | <p>To include the effect of exercise.
 <i>M0.1, M0.4, M1.2, M1.3, M1.11, M2.3, M2.4, M3.1, M3.2</i>
 PAG10, PAG11
 HSW3, HSW4, HSW5, HSW6</p> |
| <p>(f) the effect of heart rate on cardiac output</p> | <p>To include calculations based on heart rate and stroke volume.
 <i>cardiac output = heart rate × stroke volume</i>
 <i>M0.1, M0.4, M1.2, M1.3, M1.11, M2.3, M2.4, M3.1, M3.2</i>
 PAG10, PAG11
 HSW3, HSW4, HSW5, HSW6</p> |
| <p>(g) the measurement and interpretation of pulse rate, to include the generation of primary data and the use of secondary data</p> | <p><i>M0.1, M0.4, M1.2, M1.3, M1.10, M1.11, M3.1, M3.2</i>
 PAG10, PAG11
 HSW3, HSW4, HSW5, HSW6</p> |
| <p>(h) the use and interpretation of an electrocardiogram (ECG)</p> | <p>To include tachycardia, bradycardia, S-T elevation and fibrillation.
 <i>M3.1</i>
 HSW3, HSW5</p> |
| <p>(i) the emergency treatment given to a person suffering a suspected heart attack or cardiac arrest.</p> | <p>To include the first-aid treatment for a heart attack and cardiac arrest
 AND
 the use of a defibrillator following cardiac arrest.</p> |